Federated Full-Envelope LPV Identification from Incomplete Local Coverage

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***Abstract*—Individual vehicles may supply calibration data from only a narrow speed region, although the fleet collectively covers the operating interval. Pooling those records can recover missing scheduling information but also mix incompatible vehicle dynamics. We investigate compatible federated identification of linear-parameter-varying (LPV) maps under this joint coverage and heterogeneity limitation. With available lateral states, clients identify local matrices and exchange additive regression statistics and assignment scores to learn shared group maps. A nonlinear vehicle benchmark separates scheduling-rank deficiency from model mismatch and finite-data error. Across ten label-free fleets, grouped fitting reduces mean paired full-envelope matrix error by 56.20% relative to development-tuned Smooth Local and 23.74% relative to Global. At locally unobserved speeds, mean error is 3.527%, compared with 9.458% for Smooth Local and 4.388% for Global. The distributed implementation reproduces centralized grouped fitting. All 18 selected groups have full-rank scheduling designs. Separate scheduled-LQI tests provide modest supporting tracking gains. The validated setting is controlled equilibrium-centered calibration with available states, exact current regressors, and noisy next-state targets.**

***Index Terms*—federated identification, LPV systems, incomplete operating coverage, heterogeneous vehicles**

I. Introduction

Vehicles with speed-dependent and heterogeneous dynamics may observe only narrow speed regions. LPV identification then requires complementary scheduling information while avoiding bias from pooling incompatible dynamics.

Basis-dependent LPV identification requires scheduling excitation [1], [2]. Local approaches identify fixed-operating-point models before constructing a scheduling-dependent map [3]. Federated learning retains records locally [4]. For heterogeneous systems, FedSysID quantifies the tradeoff between collaboration and model bias [5]. IFCA alternates client assignment and cluster-model updates [6]. Clustered SysID applies alternating assignment and estimation to LTI systems with identical dynamics within each true cluster, providing conditional recovery and sample-complexity guarantees [7]. IC-SYSID adds incremental cluster formation and stability promotion, including a vehicle-fleet study [8]. Our previous vehicle study uses uncertainty-aware grouping for controller design [9].

Here each client's fixed speed support leaves its LPV scheduling design deficient. Repeating transitions at those speeds reduces noise but cannot identify the missing coefficient directions. Compatible pooling can supply them, conditional on group coverage, while within-group variation remains. The problem couples scheduling-information deficiency with heterogeneity bias. With common physical states, we specialize the established alternating framework to LPV matrix regression. Its guarantees do not establish recovery for our fixed deficient scheduling supports and approximate shared maps. Figure 1 summarizes the procedure.

The contribution is a distributed LPV construction and controlled study of joint coverage and heterogeneity. First, blocked-speed validation separates representation error from information deficiency. Second, clients fit group maps through additive statistics and assignment scores without full local coefficient maps or family labels, followed by a group-rank audit. Third, paired fleet comparisons separate coverage and compatibility benefits at recipient-unobserved grid speeds. Centralized–federated agreement verifies implementation. Nonlinear scheduled-LQI tests check downstream usability.

II. Identification Problem

A. Available states and incomplete speed coverage

Client i has lateral states $x = [\beta, r]^\top$, steering input u, and longitudinal speed v. The benchmark uses radians for β, u and rad/s for r. At a fixed cornering equilibrium, local perturbations satisfy approximately

$$\Delta x_{i,k+1} = A_i(v)\Delta x_{i,k} + B_i(v)\Delta u_{i,k} + e_{i,k}, \qquad (1)$$

where A_i, B_i are discrete-time matrices and $e_{i,k}$ includes nonlinear approximation error and next-state observation noise. Define $m_i(v) = [A_{11}, A_{12}, A_{21}, A_{22}, B_1, B_2]$. A group g_i, unknown during fitting, approximates the client map as

$$m_i(v) \approx \phi(v)^\top \Theta_{g_i}, \qquad \Theta_g \in \mathbb{R}^{L\times 6}. \qquad (2)$$

Within-group variation remains. Each client provides matrix estimates at speeds in $\mathcal{V}_i \subset \mathcal{V} = [10, 30]$ m/s. "Full envelope" means this speed interval along the prescribed cornering curve, rather than an arbitrary nonlinear state/input domain.

B. Fixed representation and local estimation

The scheduling span is

$$b(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v-20)/10. \quad (3)$$

Let the rows of B_{ref} be $b(v)^\top$ on a fixed 401-point uniform 10–30 m/s grid, with thin QR factorization $B_{\mathrm{ref}} = QR$. The implemented row basis is $\phi(v)^\top = b(v)^\top R^{-1}$. This invertible transform changes conditioning, not the represented span. Development-only blocked-speed validation selects $L = 7$ for cornering, while the straight-line negative control retains $[1, v^{-1}, v^{-2}]$. No control comparison selects the representation.

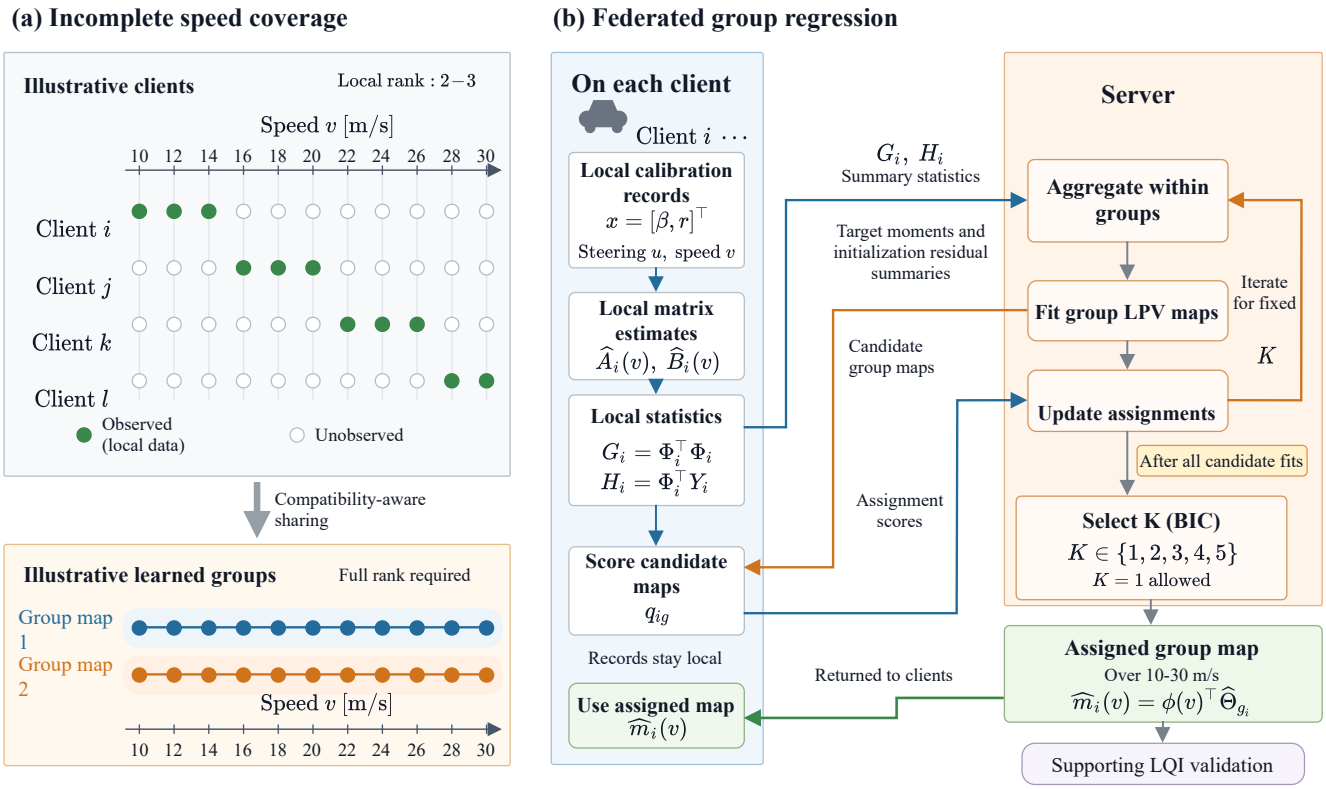


Fig. 1. Overview of compatible federated LPV identification. (a) Illustrative clients observe narrow speed blocks. Their scheduling designs have rank two or three for a seven-function basis; compatible sharing can supply missing directions if the pooled group design has rank seven. The displayed groups are illustrative. (b) Clients retain calibration records and local matrix targets, exchange additive statistics, summaries, and assignment scores, and receive fitted group maps. Fitting and assignment iterate for each candidate K, followed by BIC-based selection. Available states and equilibrium-centered calibration are assumed. The envelope is the 10–30 m/s interval along the prescribed cornering curve; LQI provides a supporting usability check.

At each observed speed, a client stacks 36 independent transitions. With rows $[\Delta x_k^\top, \Delta u_k]$ in Z and next-state rows in X_+, it estimates

$$\widehat{C} = \arg\min_C \|X_+ - ZC\|_F^2, \qquad [\widehat{A}, \widehat{B}] = \widehat{C}^\top. \tag{4}$$

Current state/input regressors are exact, and only next-state targets receive noise. These estimates, with their speeds, are the group-regression targets. Vehicle parameters, exact evaluation matrices, and family labels do not enter that fitting or assignment interface.

The generator computes each centering equilibrium from simulator parameters. Thus available states alone do not imply identification from ordinary driving records. A controlled experiment could instead measure an operating-point center, but that alternative is not validated here. The same clients contribute calibration records and receive the maps. Evaluation at additional speeds tests operating-condition generalization; fresh fleets repeat the whole procedure rather than testing a frozen library on wholly new recipients.

III. Federated Compatible Regression

A. Additive fitting and conditional identifiability

Client i forms $\Phi_i \in \mathbb{R}^{n_i \times L}$ and matrix-entry targets $Y_i \in \mathbb{R}^{n_i \times 6}$. It contributes $G_i = \Phi_i^\top \Phi_i$ and $H_i = \Phi_i^\top Y_i$. For a fixed partition, the group fit is

$$\widehat{\Theta}_g = \left(\sum_{i:g_i=g} G_i + (\lambda + \epsilon) I \right)^{-1} \sum_{i:g_i=g} H_i, \tag{5}$$

with $\lambda = 10^{-8}$ and numerical diagonal $\epsilon = 10^{-12}$. The same unweighted ridge objective and basis are used in centralized and federated grouped fitting. The matrix estimates have equal weight; local estimation covariance is not propagated. Unequal noise or excitation could therefore motivate weighting, but is outside the present comparison.

Proposition 1: For a fixed partition and exact shared-map targets $Y_i = \Phi_i \Theta_g$, the unregularized coefficients are unique if and only if the vertically stacked group design has rank L, equivalently $\operatorname{rank}(\sum_{i:g_i=g} G_i) = L$.

Justification: The aggregate Gram matrix is positive definite exactly when the stacked design has no nonzero null direction. Otherwise, adding a null-direction coefficient matrix leaves all observed predictions unchanged. Additivity also makes the centralized and distributed solutions identical under the same numerical conventions.

This statement is conditional on a partition and model class. It neither establishes correct group discovery nor removes

TABLE I
FEDERATED GROUP-REGRESSION PROCEDURE.

Inputs: Local speed/matrix pairs, fixed basis, candidate $K = 1, \ldots, 5$, minimum group size five.
1. Aggregate local statistics and target moments. Fit Global, broadcast it, and collect residual summaries.
2. For each K, use one initialization for $K = 1$ or twelve seeded K-means initializations otherwise.
3. Alternate group-statistic solves, map broadcasts, local score returns, and hard assignments. Stop at unchanged assignments or 30 iterations. Discard initializations with groups smaller than five.
4. Refit the final admissible partition if needed. Retain the lowest-BIC initialization for each K, then the lowest-BIC K.
Outputs: Group maps and client assignments, including $K = 1$.

pooling bias. Ridge gives a numerical answer in deficient groups but does not restore information.

B. Assignment and model selection

The server derives a common entry scale from additive counts, sums, and squared sums of Y_i. For entry j, let s_j be its pooled standard deviation, floored at 0.05 times the median standard deviation, and set $D = \mathrm{diag}(s_1, \ldots, s_6)$. Clients return scores

$$q_{ig} = \|(Y_i - \Phi_i \widehat{\Theta}_g) D^{-1}\|_F^2, \qquad g_i \leftarrow \arg\min_g q_{ig}. \tag{6}$$

The common scaling balances entries during assignment. Fitting remains entry-wise ridge regression. Initial groups come from K-means on standardized six-entry client mean residuals around the Global fit. The server receives those summaries, not the local matrices themselves.

The model-selection score uses the normalized assignment residuals:

$$\mathrm{BIC} = n \log\big(\max\{\mathrm{SSE}/n, 10^{-12}\}\big) + p_K \log n, \tag{7}$$

where $\mathrm{SSE} = \sum_i q_{i,g_i}$, $n = 6\sum_i n_i$, and $p_K = 6LK + K - 1$. Ties choose the first group index. This is a heuristic selection criterion for finite noisy matrix targets and within-group variation. It does not certify a likelihood model, global convergence, or physical-family recovery. Minimum group size does not imply full scheduling rank, so group fit and group information must be assessed separately.

C. Communication

For $L = 7$, each local Gram has 28 distinct entries and its right-hand side has 42. Target moments, initialization summaries, group tags, assignment scores, and model broadcasts also enter the accounting. Including all candidate orders and restarts, mean label-free traffic is 2.243 MB per fleet at eight bytes per scalar. This exhaustive search is not shown to be communication-efficient. The emulator retains raw records locally, but informative statistics and scores are shared. No secure aggregation or differential privacy is implemented.

TABLE II
VEHICLE CENTERS USED ONLY TO GENERATE THE FLEET. ALL CLIENTS HAVE $l_f = 1.2$ M AND $l_r = 1.6$ M.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

IV. BENCHMARK AND EVALUATION

A. Nonlinear vehicles and calibration records

The plant is a two-state single-track model with nonlinear tanh tires:

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \tag{8}$$

$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \tag{9}$$

$$F_a = \mu F_{za} \tanh(C_a \alpha_a/(\mu F_{za})), \quad a \in \{f, r\}. \tag{10}$$

Here $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, $F_{zr} = mgl_f/(l_f + l_r)$, $g = 9.81$ m/s^2, and $\mu = 0.9$. Each fleet has ten clients from each center in Table II. Mass and inertia receive independent uniform multiplicative scatter of $\pm 2.5\%$ and axle stiffnesses $\pm 5\%$. Geometry remains fixed. Family labels are withheld from primary block allocation and fitting.

Calibration follows curvature 0.005 m^{-1} and uses RK4 integration with sampling time 0.01 s. Zero-mean independent Gaussian perturbation standard deviations are approximately 0.25° for β, 2°/s for r, and 0.30° for steering. Added next-state noise has standard deviations 0.01° and 0.03°/s. These are independently initialized one-step records, not successive samples from a driving trajectory.

The four local speed blocks are $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s. Globally balanced random allocation assigns them to eight, eight, seven, and seven clients, respectively, without consulting family labels. Primary fleet seeds are 371–380. Development representation selection uses separate seeds 301–310, contiguous blocked-speed validation, a one-standard-error rule, and a practical relative-error floor of 10^{-4}.

B. Endpoints, averaging, and baselines

Reference matrices $m_i^\star(v)$ are numerical continuous-time Jacobians at the prescribed equilibrium, discretized by an augmented matrix exponential. They are used for development representation and local-penalty selection and evaluation, not finite-data group fitting. The primary relative error is

$$e_i(v) = \frac{\|\widehat{m}_i(v) - m_i^\star(v)\|_2}{\|m_i^\star(v)\|_2}, \qquad E_s = \frac{1}{30 \cdot 11} \sum_i \sum_{v \in \mathcal{V}_{\mathrm{grid}}} e_i(v). \tag{11}$$

where $\mathcal{V}_{\mathrm{grid}} = \{10, 12, \ldots, 30\}$ m/s. The metric uses the stated physical coordinates without entry normalization. It is a fixed-coordinate benchmark error, not a coordinate-invariant measure

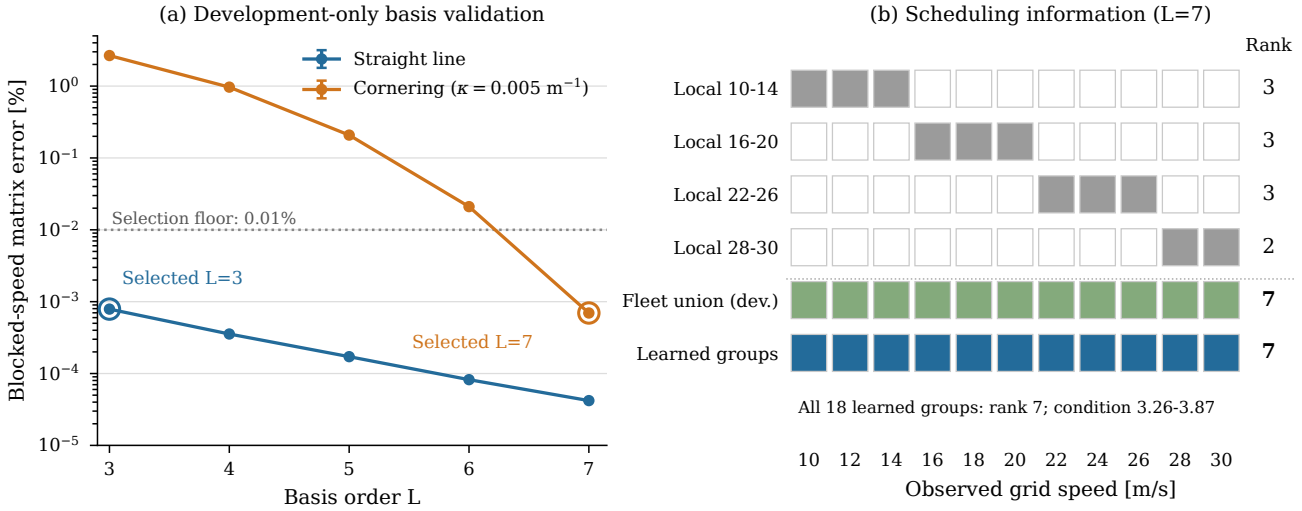


Fig. 2. Scheduling representation and information coverage. (a) Mean blocked-speed matrix error across ten development fleets; error bars denote one standard error. Rings mark the selected orders, and the dotted line is the 0.01% practical selection floor. (b) Filled cells mark observed grid speeds for the seven-function basis. The development union and all 18 audited primary groups have rank seven, unlike individual clients. Development union and learned-group coverage are distinct checks; full rank does not remove within-group bias.

or a tracking-performance surrogate. Observed/unobserved-speed summaries average only their respective client-speed pairs within each fleet. Reported errors average E_s over ten independent fleets. For method M and baseline B, reported gains average $100(1 - E_s^M/E_s^B)$ over paired fleets, rather than taking the ratio of grand mean errors. Confidence intervals resample those ten fleet units 20,000 times.

Restricted Local uses $L = 3$, ridge 10^{-8}, and each client's records. Global and both grouped methods use $L = 7$ and the same fleet records. These baselines separate complementary coverage, compatibility selection, and distributed implementation. Prior-centered Local uses $L = 7$ with ridge toward its own $L = 3$ fit expressed in the richer basis. Leave-one-speed-out validation selects the penalty from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}\}$; clients with only two speeds use the fixed middle candidate 10^{-2}. This diagnostic retains the restricted prior in local null directions, apart from numerical shrinkage.

Smooth Local instead penalizes function curvature in the same $L = 7$ span:

$$\widehat{\Theta}_i^{\mathrm{S}} = \arg\min_{\Theta} \frac{\|Y_i - \Phi_i\Theta\|_F^2}{n_i} + \alpha\|Q_2\Theta\|_F^2 + \eta\|\Theta\|_F^2. \quad (12)$$

Here $\|Q_2\Theta\|_F^2$ uses 401-point trapezoidal quadrature for $\frac{1}{2}\int_{-1}^{1} \|d^2[\phi(v(z))^\top\Theta]/dz^2\|_2^2\, dz$; $\eta = 10^{-16}$ is numerical ridge. Affine functions are unpenalized; there is no $L = 3$ coefficient prior. A common α is selected from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}, 10^{4}, 10^{6}\}$ on development fleets 321–330 using unobserved-speed matrix error. The largest penalty within one fleet standard error of the best mean is chosen: 10^{6}. Reference Jacobians serve this offline tuning; recipient fitting uses only its own noisy targets. This matched check and the group-information audit reuse the original ten fleets and calibration protocol, with reconstructed parent errors verified against saved client-speed results. They add no independent confirmation. A full-coverage recipient reference outperforms family pooling at approximately equal transition budget, bounding the benefit to missing local access rather than superiority at equal recipient information.

V. Identification Results

A. Why complementary coverage is needed

At curvature 0.005 m^{-1}, development blocked-speed matrix error falls from 2.6607% at order three to 0.0007% at order seven. Figure 2(a) contrasts this basis requirement with the straight-line case, which retains order three. For seven coefficients, the local blocks have ranks three, three, three, and two; the development union has rank seven and condition 3.406 (Fig. 2(b)). Complementary coverage can therefore supply missing directions, conditional on the learned partition.

With finite transitions and known compatibility, oracle-family pooling achieves 1.816% unobserved-speed error, versus 66.121% for Local and 4.296% for Global on separate development fleets. The prior-centered diagnostic changes Local negligibly there; this does not test a distinct extrapolation prior. These oracle results motivate the mechanism; the primary evaluation below withholds labels.

B. Primary label-free results

Table III separates the full-envelope endpoint from its observed and unobserved regions. Mean paired gains over Local and Global are 93.41% and 23.74%, respectively. The Global-relative 95% interval is $[15.39, 31.04]$%, with improvement in eight of ten fleets. Two groups are selected in eight fleets and one in two. Recipient-only fits are more accurate at their own observed speeds, whereas sharing substantially reduces Restricted Local's unsupported extrapolation error. Grouped sharing additionally reduces Global's pooling error. Figure 3 locates these changes relative to each client block. Figure 4(a)

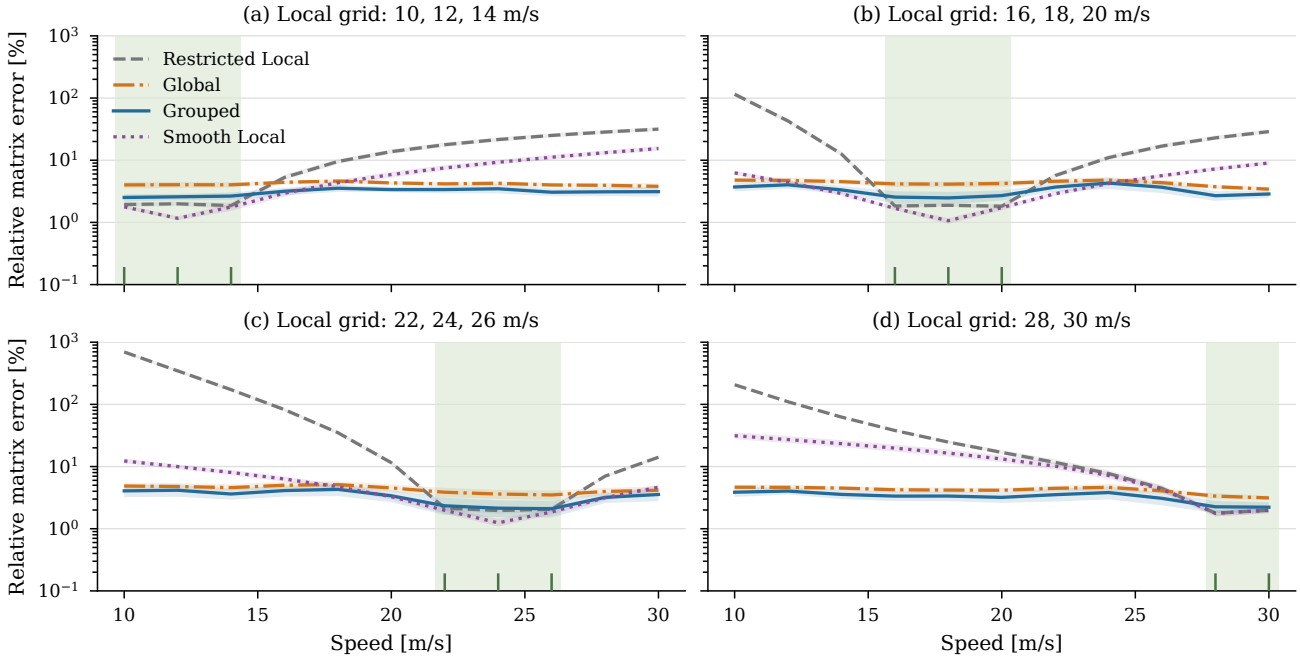


Fig. 3. Primary label-free matrix error by local speed block. Light shading marks each observed block; short green marks identify its observed grid speeds. Curves average clients within block and fleet, then across ten fleets; bands are pointwise 95% fleet-bootstrap intervals from 20,000 draws. The logarithmic axis retains Restricted Local extrapolation failure. Smooth Local uses a development-selected curvature penalty and the same frozen recipient targets. These curves and Fig. 4 reuse the primary records, providing no additional independent confirmation.

shows the paired fleet errors, including the two $K = 1$ ties. Aggregate improvement does not extend to every client: 169 instances improve, 60 are numerical ties, and 71 have larger unobserved-speed error than Global. Figure 4(b) shows this tradeoff. Its 300 client instances are nested within ten fleets and are descriptive, rather than independent replicates for confidence intervals.

Prior-centered Local changes unobserved-speed error by only 0.00016 percentage points. Its prior's observed-target relative residual has median 0.0020% and maximum 0.0708% across 300 clients. Small residuals drive small corrections, while unsupported directions retain the prior. This diagnostic provides little independent evidence about local regularization.

Smooth Local substantially reduces Local's extrapolation error and has the lowest observed-speed error (Table III). Grouped fitting still improves full-envelope error by 56.20%, with paired fleet-bootstrap interval $[49.69, 61.81]\%$, and unobserved-speed error by 62.33%, interval $[57.03, 67.08]\%$, winning in all ten fleets for both endpoints. Figure 3 shows the remaining unobserved-speed gap; other local priors remain untested.

Federated and centralized protocols select identical partitions and group counts in all 20 fleet/protocol jobs, including the separate family-stratified coverage reference. Maximum coefficient discrepancy is 3.20×10^{-14} and maximum BIC discrepancy is 1.87×10^{-9}. This is implementation verification under matched inputs.

TABLE III
MEAN MATRIX ERROR [%] ACROSS TEN LABEL-FREE FLEETS. THE TWO $L = 7$ LOCAL FITS USE DISTINCT PRIORS.

Method	Full	Observed	Unobserved
Restricted Local	49.562	1.930	65.568
Prior-centered Local	49.562	1.930	65.568
Smooth Local	7.489	1.630	9.458
Global	4.250	3.840	4.388
Centralized groups	3.249	2.423	3.527
Federated groups	3.249	2.423	3.527

C. Scheduling information in selected groups

All 18 selected groups contain observations at all eleven grid speeds and have scheduling-design rank seven. Their stacked design condition numbers range from 3.26 to 3.87 in the fixed QR basis, as summarized in Fig. 2(b). The learned partitions satisfy Proposition 1's scheduling condition for this benchmark. The reported conditioning concerns the scheduling design, rather than its Gram or the full dynamical-identification information matrix. Full rank neither removes within-group bias nor guarantees physical-family recovery.

A known-family reference loses a speed block in one fleet and produces an infeasible full-envelope controller, illustrating that compatibility labels alone do not establish information coverage.

VI. SUPPORTING LQI VALIDATION

Ten separate confirmation fleets (seeds 411–420) refit the maps and apply speed-scheduled LQI to nonlinear 12 s

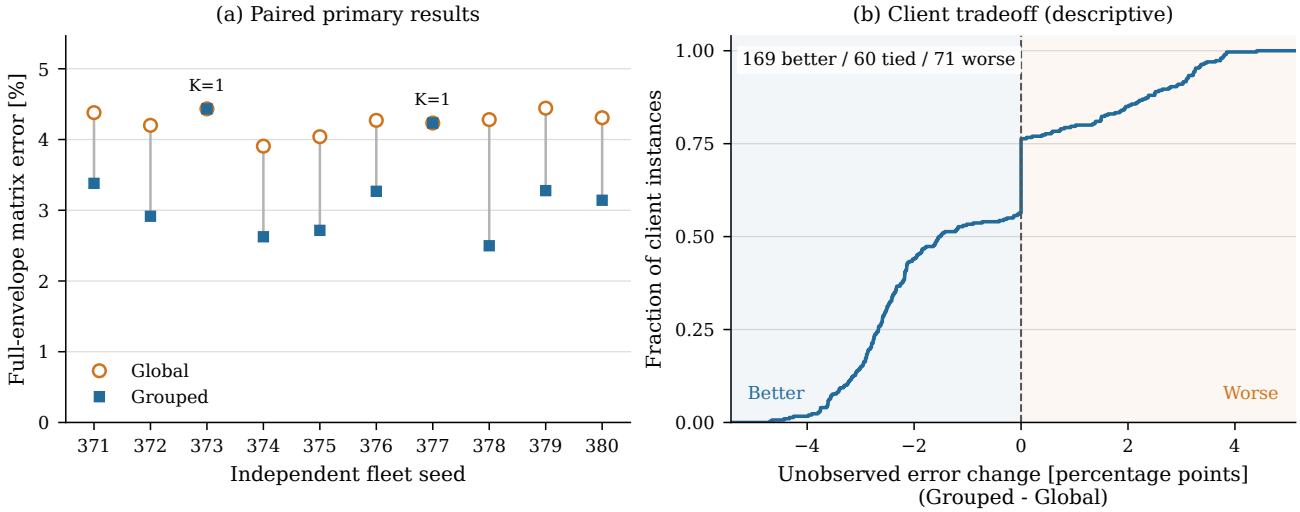


Fig. 4. Primary fleet gains and client tradeoffs. (a) Paired full-envelope errors for all ten independent fleets; connectors join methods within each fleet, and $K = 1$ marks the two ties. (b) Descriptive empirical cumulative distribution of client mean unobserved-speed error changes (Grouped minus Global) in percentage points; negative values indicate improvement. Ties use a 10^{-12} dimensionless-error tolerance. The 300 client instances are nested within ten fleets.

TABLE IV
SUPPORTING YAW TRACKING RMSE [DEGREES/S], AVERAGED ACROSS TEN CONFIRMATION FLEETS. OUTSIDE ITS OBSERVED BLOCK, LOCAL USES THE NEAREST-OBSERVED-SPEED CONTROLLER.

Method	Moderate	Hard
Restricted Local	0.0540	0.1698
Global	0.0429	0.1564
Federated groups	0.0411	0.1546
ExactLPV reference	0.0400	0.1507

maneuvers with available physical-state feedback. At the eleven grid speeds, discrete Riccati synthesis uses $Q = \mathrm{diag}(25, 50, 1000)$ for the two states and yaw-error integral, with steering weight $R = 0.15$. Gains and reference prefilters are linearly interpolated in speed. Weights were selected using only ExactLPV performance on development seeds 381–385, subject to feasibility and saturation checks. A weaker maneuver pilot on seeds 391–400 was discarded before the final confirmation. Neither grouped-versus-Global results nor confirmation fleets select weights.

The moderate maneuver smoothly varies speed $11 \to 29 \to 15$ m/s and uses an enveloped two-frequency lateral-acceleration demand with scale 4 m/s^2 and frequencies 0.35/0.72 Hz. The hard maneuver varies speed $29 \to 12 \to 28$ m/s and uses alternating Gaussian demands centered at 3.0/6.0/9.2 s, widths 0.40/0.48/0.55 s, and amplitudes $7/-7.7/5.6$ m/s^2. Yaw reference is requested acceleration divided by speed. Steering limits are 12° and 120°/s, with conditional integration, integral clipping at 0.6, and a 6° sideslip feasibility bound. These remain fixed across methods.

Mean paired tracking gains are 23.86%/8.96% against Local and 4.26%/1.21% against Global for moderate/hard demands. Global-relative intervals are $[1.95, 6.55]\%$ and $[0.47, 2.01]\%$. Absolute mean improvements are 0.001824/0.001874 degrees/s, so the control evidence supports usability with modest practical headroom. All federated trajectories satisfy the tested constraints. Maximum frozen-point spectral radius is 0.9759, which is not a scheduled or nonlinear stability certificate. Four fleets select one group, making federated and Global controllers identical there.

VII. DISCUSSION AND CONCLUSION

Complementary speed blocks supply directions absent from local regressions, while compatibility selection reduces pooling error. Grouped fitting improves full-envelope identification over Restricted Local, the distinct Smooth Local comparator, and Global, and federation reproduces centralized fitting on matched records. The figures also expose the tradeoff: recipient-only fits are more accurate at observed speeds, and some clients deteriorate relative to Global. The benefit is aggregate access across the speed interval, rather than uniform client improvement or recovery of the generating physical families.

The controlled calibration uses simulator-derived centering and exact current regressors along one cornering curve. Ordinary driving, noisy state estimates, real fleets, and frozen-library transfer remain unvalidated. Group selection is heuristic: full rank is empirical, minimum size does not guarantee it, and exhaustive search incurs measurable communication.

Compatible federated regression therefore supplies useful LPV maps across the speed interval under the stated assumptions. Nonlinear scheduled-LQI tests support usability with modest tracking gains. Smooth Local has only been compared on matrix error; its downstream control performance remains untested.

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